

g) develop the winding diagram wave wound DC armature with 4-pole 18 conductors

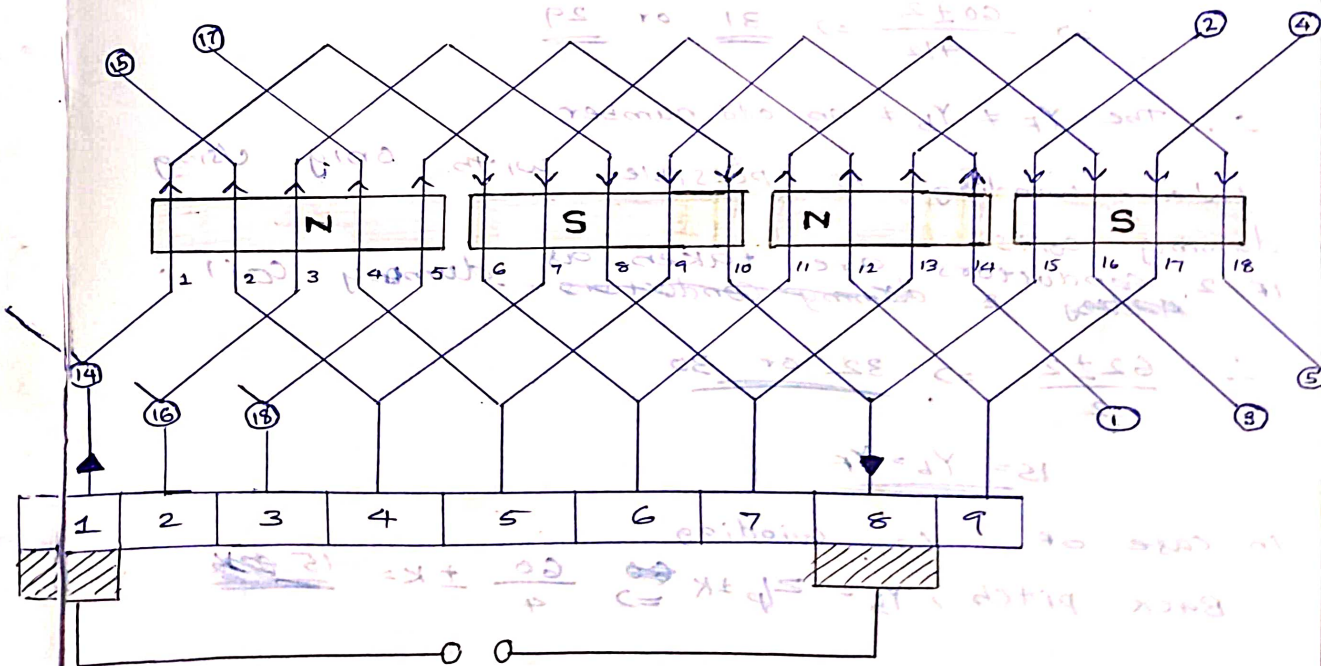
$$Y_w = \frac{Z \pm 2}{P/2} \Rightarrow \frac{18 \pm 2}{4/2} =$$

10 or 8

$$5 = Y_f = Y_b$$

Winding table

Back Connection	Front Connection
$1+5 = 6$	$6+5 = 11$
$11+5 = 16$	$16+5 = 3$
$3+5 = 8$	$8+5 = 13$
$13+5 = 18$	$18+5 = 5$
$5+5 = 10$	$10+5 = 15$
$15+5 = 2$	$2+5 = 7$
$7+5 = 12$	$12+5 = 17$
$17+5 = 4$	$4+5 = 9$
$9+5 = 14$	$14+5 = 1$



6. The number of coils i.e. N_C can be found from the relation.

$$N_C = \frac{PY_A \pm 2}{2}$$

This relation has been found by rearranging the relation given in (5) above.

7. It is obvious from (5) that for a wave winding, the number of armature conductors with 2 either added or subtracted must be a multiple of the number of poles of the generator. This restriction eliminates many even numbers which are unsuitable for this winding.
8. The number of armature parallel paths = $2m$ where m is the multiplicity of the winding.

Example 26.2. Draw a developed diagram of a simplex 2-layer wave-winding for a 4-pole d.c. generator with 30 armature conductors. Hence, point out the characteristics of a simple wave winding.

(Elect. Engg-I, Nagpur Univ. 1991)

Solution. Here, $Y_A = \frac{30 \pm 2}{4} = 8^* \text{ or } 7$. Taking $Y_A = 7$, we have $Y_B = Y_F = 7$

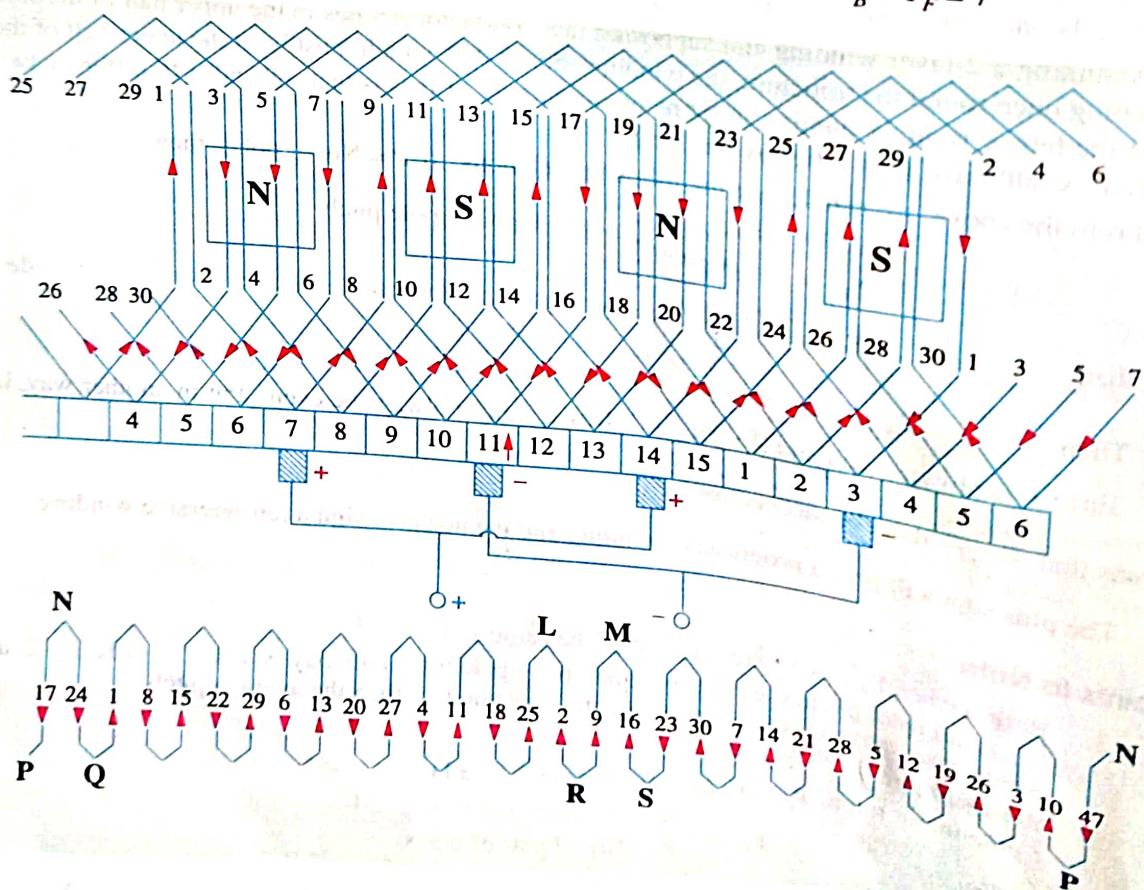


Fig. 26.36

As shown in Fig. 26.36 and 26.37, conductor No. 5 is taken to conductor No. 5 + 7 = 12 at the back and is joined to commutator segment 5 at the front. Next, the conductor No. 12 is joined to commutator segment 5 + 7 = 12 ($\because Y_C = 7$) to which is joined conductor No. 12 + 7 = 19. Continuing this way, we come back to conductor No. 5 from where we started. Hence, the winding closes upon itself.

* If we take 8, then the pitches would be : $Y_B = 9$ and $Y_F = 7$ or $Y_B = 7$ and $Y_F = 9$. Incidentally, if $Y_A = Y_C$ is taken as 7, armature will rotate in one direction and if $Y_C = 8$, it will rotate in the opposite direction.

The simple winding table is as under :

Back Connections

$$1 \text{ to } (1 + 7) = 8$$

$$15 \text{ to } (15 + 7) = 22$$

$$29 \text{ to } (29 + 7) = 36 = (36 - 30) = 6$$

$$13 \text{ to } (13 + 7) = 20$$

$$27 \text{ to } (27 + 7) = 34 = (34 - 30) = 4$$

$$11 \text{ to } (11 + 7) = 18$$

$$25 \text{ to } (25 + 7) = 32 = (32 - 30) = 2$$

$$9 \text{ to } (9 + 7) = 16$$

$$23 \text{ to } (23 + 7) = 30$$

$$7 \text{ to } (7 + 7) = 14$$

$$21 \text{ to } (21 + 7) = 28$$

$$5 \text{ to } (5 + 7) = 12$$

$$19 \text{ to } (19 + 7) = 26$$

$$3 \text{ to } (3 + 7) = 10$$

$$17 \text{ to } (17 + 7) = 24$$

Front Connections

$$8 \text{ to } (8 + 7) = 15$$

$$22 \text{ to } (22 + 7) = 29$$

$$6 \text{ to } (6 + 7) = 13$$

$$20 \text{ to } (20 + 7) = 27$$

$$4 \text{ to } (4 + 7) = 11$$

$$18 \text{ to } (18 + 7) = 25$$

$$2 \text{ to } (2 + 7) = 9$$

$$16 \text{ to } (16 + 7) = 23$$

$$30 \text{ to } (30 + 7) = 37 = (37 - 30) = 7$$

$$14 \text{ to } (14 + 7) = 21$$

$$28 \text{ to } (28 + 7) = 35 = (35 - 30) = 5$$

$$12 \text{ to } (12 + 7) = 19$$

$$26 \text{ to } (26 + 7) = 33 = (33 - 30) = 3$$

$$10 \text{ to } (10 + 7) = 17$$

$$24 \text{ to } (24 + 7) = 31 = (31 - 30) = 1$$

Since we come back to the conductor No. 1 from where we started, the winding gets closed at this

stage